Plan of Attack

02 December 2024 18:41

1. We will build a simple neural network
2. Train it on a real world dataset
3. Will mimic the PyTorch workflow
4. Will have a lot of manual elements
5. End result is not important

3 videos →

Intro

Tensors

Computation + Autograd

Breast cancer

Dataset → NN

↓

Training pipeline → foundation

Code flow

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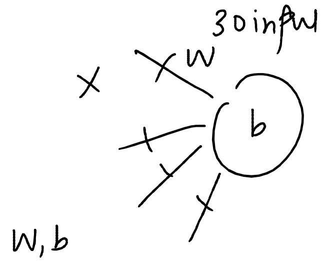
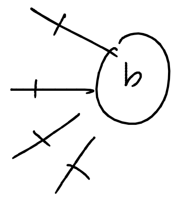
1. Load the dataset
2. Basic preprocessing
3. Training Process
    a. Create the model
    b. Forward pass
    c. Loss calculation
    d. Backprop
    e. Parameters update
4. Model evaluation → accuracy





$$z = wx + b$$

$$\sigma(z)$$

$$W_{new} = w_{old} - lr \frac{\partial L}{\partial w}$$

$$b_{new} = b_{old} - \eta \frac{\partial L}{\partial b}$$

Next steps

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